

Impinj Speedway
Revolution R420

Impinj Speedway R420 is a high-performance 4-port fixed UHF RFID reader with the largest installed base across diverse applications. With patented Autopilot capability for automatic performance optimization, GS1/EPCglobal UHF Gen2 (ISO 18000-6C) compliance, and IEC IP52 environmental sealing, it delivers proven reliability for enterprise-grade item visibility solutions.



UHF RFID

Air Interface Protocol	GS1/EPCglobal UHF Gen2 (ISO 18000-6C) or RAIN RFID
Operating Frequencies (R420)	FCC: 902-928 MHz; EU1: 865-868 MHz; EU2: 915-921 MHz; GX1/GX2/GX3; JP2
Operating Frequencies (R220)	FCC: 902-928 MHz; EU1: 865-868 MHz; GX1: 902-928 MHz; GX2: 902-925 MHz
Transmit Power (R420)	FCC/GX1/GX2/GX3: 32.5 AC/31.5 PoE; EU1: 31.5 AC/30.0 PoE; EU2: 33.0 AC/33.0 PoE+
Antenna Ports	4 (R420); 2 (R220)
Maximum Read Rate	1100 tags/s (R420); 200 tags/s (R220)
Receive Sensitivity (max)	-84 dBm
Return Loss (min)	10 dB
Antenna Impedance	50 ohm
Antenna Hub Support	Yes (up to 32 read zones with 4 hubs on R420)

CONNECTIVITY

Network Connectivity	10/100BASE-T Ethernet
USB Ports	1 device, 1 host
Management Console Port (RS-232)	RJ-45
GPIO Serial Port	Yes
Power Sources (R420)	AC-DC power supply; IEEE 802.3af PoE (all models except EU2); 802.3at PoE+ (EU2)

MECHANICAL

Dimensions (H x W x D)	7.5 x 6.9 x 1.2 in (19 x 17.5 x 3 cm)
Weight	1.5 lb (0.7 kg)
Factory Reset Button	Yes

ENVIRONMENTAL

Operating Temperature	-4 °F to +122 °F (-20 °C to 50 °C)
Storage Temperature	-4 °F to 140 °F (-20 °C to 60 °C)
Humidity	5% to 95% non-condensing
Environmental Sealing	IEC IP52
Environmental Air Handling	NEC section 300-22(a)
Shock and Vibration	MIL-STD-810G
RoHS Compliant	Yes
Reliability	Enterprise Grade

SOFTWARE

Network Protocols	SSH, HTTP, HTTPS, NTP, DHCP, SFTP, mDNS
Networking Stack	IPv4, IPv6
Management	SNMPv1, v2, v3
Security	802.1x (Port Security), TLS 1.2 for Secure LLRP
Host Interface	LLRP with Impinj Extensions
Custom Application Partition	Yes
Development Libraries	Octane SDK (.NET, Java), LTK (C, C++, .NET, Java), ETK (C, C++)
Software supported	Impinj ItemTest, Impinj ItemSense, Impinj Speedway Connect

FEATURE HIGHLIGHTS

- Patented Autopilot for automatic performance optimization
- High read rates resistant to RF noise and interference
- Supports up to 32 antennas with antenna hubs (R420)
- Power over Ethernet (PoE) support for simplified deployment
- Enterprise-grade reliability and security
- Industry-leading proven reliability and largest install base

PRODUCT GROUP

Category	RFID Reader
Product Group	Fixed UHF RFID Reader